Co-author update: May-September 2026 revision and the human-impact proxy decision

An update from Ondrej to Vivian and Suzette

2026-09-27

Vivian and Suzette,

I have been working through the reviewers' comments and revising the analyses and manuscript since May. There is quite a lot to catch up on, so I wanted to walk you through what I changed, why I changed it, and what the results now allow us to say. I will follow the order of the revision, starting with the pollen data and response trajectories, then the variation partitioning and sensitivity checks, and finally the changes to the manuscript.

Most of this work is already incorporated in the revised manuscript and reviewer response, which I refer to here as R1. The main analyses still use the summed probability distribution of archaeological radiocarbon dates (SPD) as the human predictor. Towards the end, I will show you a later analysis that raises a question we need to resolve together before I go further with the paper. I have kept that exploratory work separate from R1.

1 How the revision developed

The table below gives you a map of the work from May to September. Some steps address the reviewers' concerns; others came from checking our analyses more closely and finding things I needed to correct or explain better.

Table 1: Revision map from May to September 2026.

Period	Change	Why it matters
May-June	Pollen-count extraction and retention diagnostics	Separated data coverage from model robustness and documented the precision-coverage trade-off.
June-July	Ten PAP trajectories refitted and exported	Made response histories visible before attribution and prevented long sequences from dominating summaries.
June-July	PAP correlation and reduced-response analysis	Tested whether overlapping assemblage properties determined H1.
July-August	HVarPart extraction corrected and model fit separated	Replaced the old ratio with commonality, signed components, display compositions, and adjusted R-squared.
August	Time and spatial controls added to H1	Made the spatial and temporal comparisons reciprocal and explicit about dependence.
August-September	Dependence, thinning, omission, radius, and event checks	Tested sampling structure and alternative definitions on matched subsets.
September	R1 manuscript and evidence registry rebuilt	Aligned scope, terminology, figures, replies, and reproducible public evidence.
Late September	Exploratory follow-up on human-impact proxies	Raised the question of how to represent human impact; discussed at the end and not yet in R1.

As you read, it helps to keep three parts of the work in mind:

1. **The revised manuscript analyses:** the results and figures I have already incorporated in R1.
2. **The supporting checks:** sensitivity and validation analyses that help us understand the limits of those results.
3. **The later exploratory work:** the additional human-proxy comparison at the end, which I have not yet incorporated in the manuscript or response.

Alongside the scientific work, I reorganised the scripts into registered, restartable pipelines. The outputs now have defined analysis profiles, source hashes, and a registry linking manuscript claims to their evidence. This does not change the results by itself, but it makes it much easier to trace each value and figure back to its source.

2 First, clarifying which data enter each analysis

Before getting into the individual changes, I want to clarify which sequences actually contribute to the results. The preparation pool contains 1,270 pollen sequences. The initial spatial inventory contains 1,262, but the usable subset differs among analyses because of response availability, predictors, temporal coverage, controls, and rank requirements. I made these differences explicit in the manuscript so that we no longer give the impression that every sequence contributes to every result.

Table 2: Analysis-specific populations used by the revised evidence.

Analysis	Unit	Number
Preparation pool	pollen sequences	1270
Initial spatial H1 inventory	pollen sequences	1262
Estimable initial spatial H1	pollen sequences	1156
Controlled spatial H1	pollen sequences	1098
Controlled temporal H1	continental-region-age comparisons	70
Matched SPD-radius spatial comparison	pollen sequences	999
Matched expert-event spatial comparison	pollen sequences	151

I also clarified why the main SPD comparison covers 8.5–2.0 ka BP. The younger boundary gives us a practical common interval given the archaeological coverage; it does not mean that SPD is universally invalid after 2 ka BP. The expert-event histories extend later in some regions, so I interpret those separately.

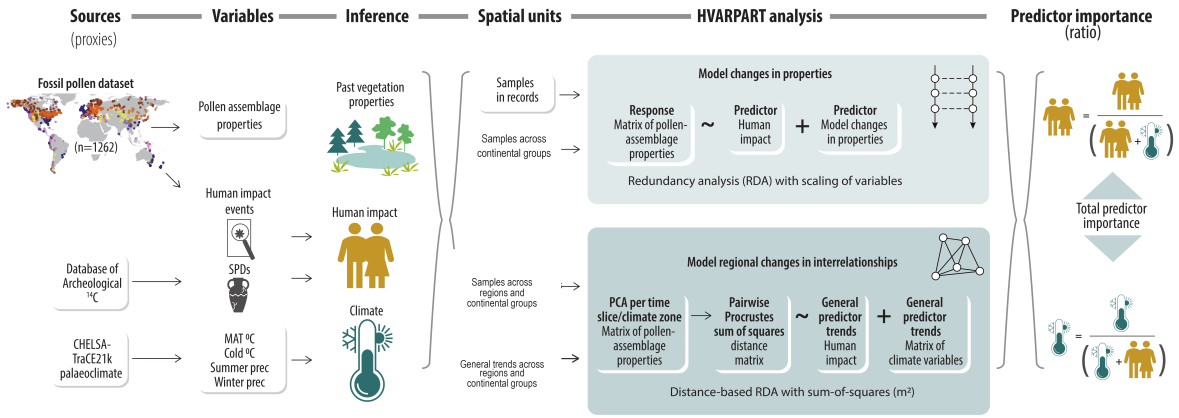


Figure 1: Current R1 study overview. The figure distinguishes the data sources, analytical scales, and the two hypotheses.

3 Checking pollen counts and the cost of stricter thresholds

One of the first things I checked was how much data we would lose by requiring higher pollen counts. The baseline profile retained 1,270 sequences. Increasing the nominal count target leaves sequences at 300 grains, at 500 grains, and at 1,000 grains. Those losses are uneven among continental regions, so a stricter threshold also changes the geographical coverage.

On that basis, I retained the baseline rule and made the precision-coverage trade-off explicit. There is an important limit to this check: I examined coverage, but did not refit H1 under the higher thresholds. We can therefore describe the data loss, but cannot claim that the H1 estimates would be unchanged.

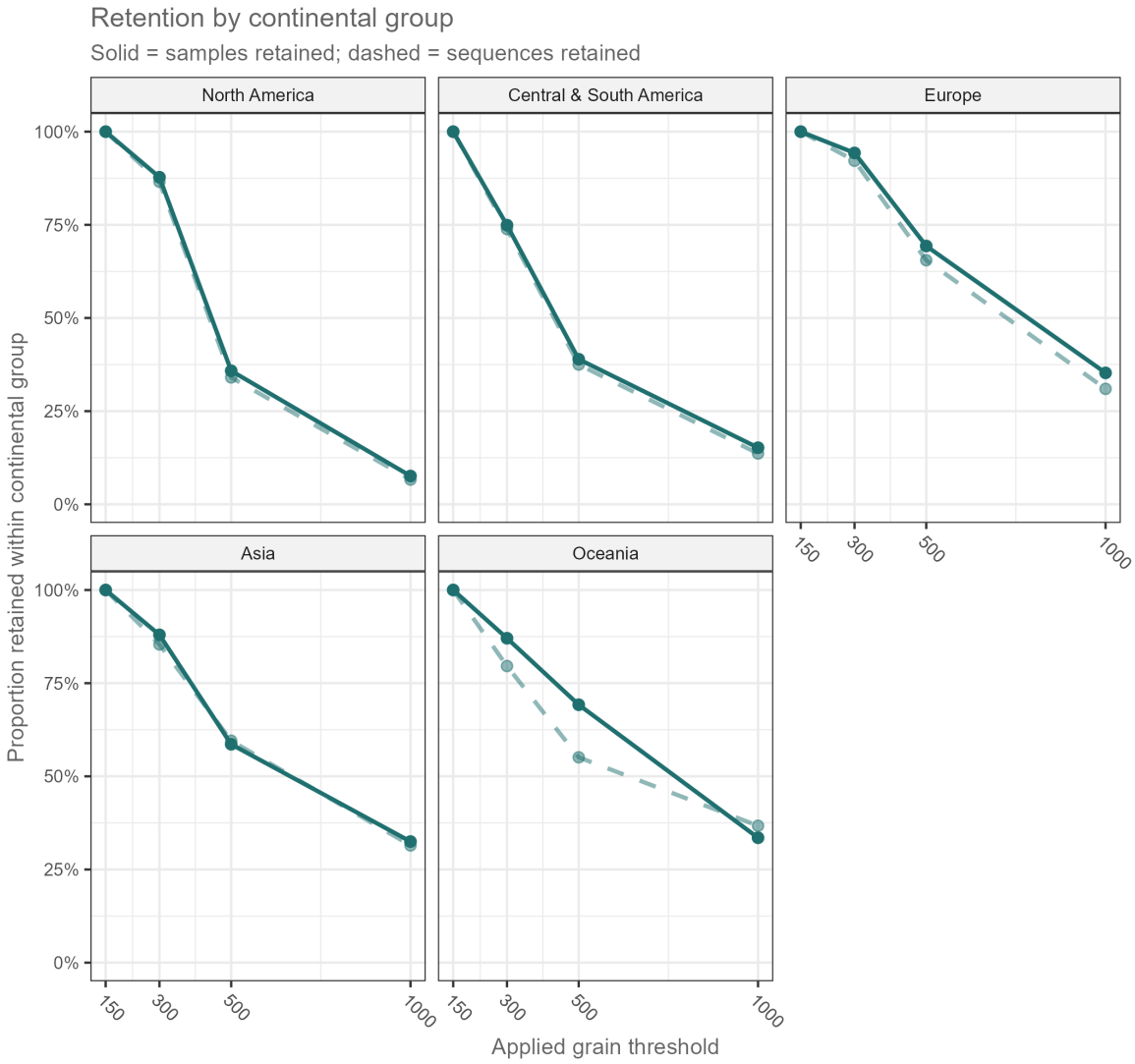


Figure 2: Pollen-count diagnostics used in R1. Retention profiles show the increasingly severe and geographically uneven data loss under higher count targets.

4 Refitting the pollen assemblage trajectories

I then revisited the response trajectories so that we could show how the pollen assemblage properties (PAPs) change before trying to attribute those changes to human activity or climate. The revised framework contains ten PAPs rather than nine, including the difference between Shannon- and Simpson-weighted diversity. I refitted all temporal hierarchical generalised additive models (hGAMs) through a reproducible request-and-export workflow. The plotted trajectories are equal-weighted summaries of sequence-level posterior predictions within observed temporal coverage, so long or densely sampled sequences do not dominate the regional curves.

What stands out to me is the heterogeneity. Richness, diversity, evenness, turnover, rates of change, and change-point densities do not move synchronously in every region. I have also made clear that gaps in the curves indicate inadequate observed coverage, rather than an absence of ecological change. The main figure shows selected trajectories, and all ten PAPs are available in the Supplementary Information.

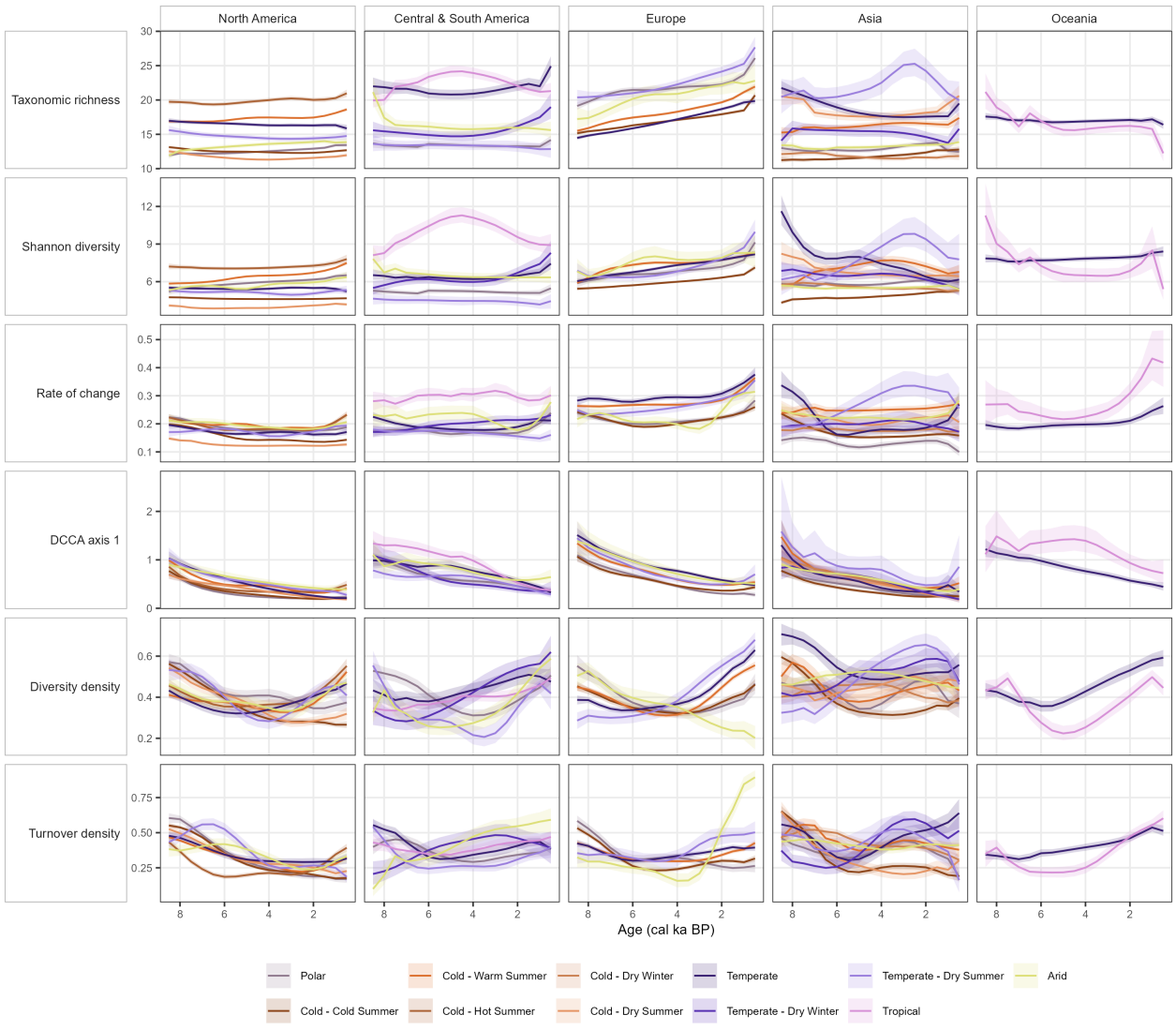


Figure 3: Revised manuscript Figure 2: selected temporal trajectories of pollen assemblage properties. The Supplementary Information shows all ten PAPs.

5 Checking whether overlapping PAPs drive the result

With those trajectories in place, I checked how much the PAPs overlap. We expect some correlation because several properties share taxa, abundance information, or mathematical components. Of 45 PAP pairs, 3 have a median absolute correlation of at least 0.8 across eligible groups; the maximum median absolute correlation is 0.96.

I therefore ran a reduced-response sensitivity that removes redundant PAPs, comparing only sequences for which both configurations can be estimated. Across 1,114 matched sequences, the aggregate climate-over-human ordering persists, although individual allocations change. This reassures me about the broad ordering in the SPD-based analysis, but it does not make the ten PAPs independent replicates or imply that every sequence gives the same estimate.

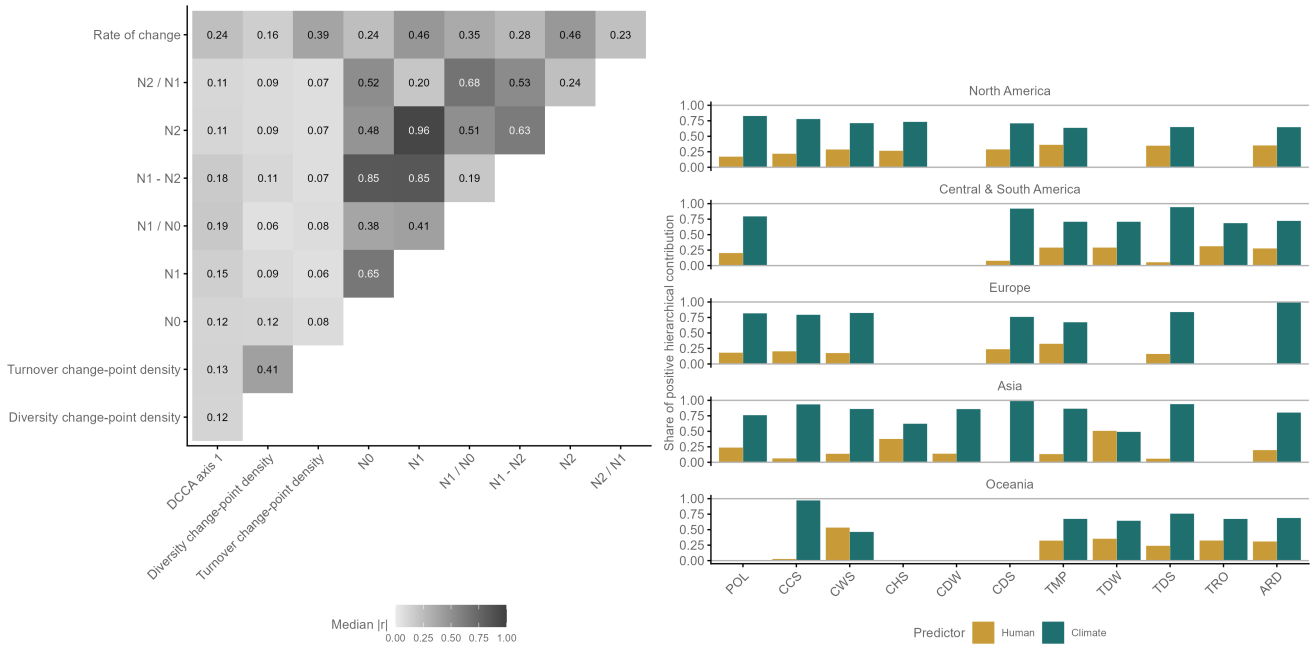


Figure 4: PAP correlation structure (left) and matched full- versus reduced-response H1 allocations (right).

6 Correcting how I extract and interpret HVarPart results

The next substantial change concerned the variation partitioning. When I revisited the original HVarPart extraction, I found that it treated relative importance too simply. I corrected the workflow to use the complete commonality decomposition, which separates unique and shared components of explained variation, and kept four quantities distinct:

- **Total adjusted (R^2):** how much multivariate response variation the full model explains.
- **Unique adjusted (R^2):** non-shared variation associated with one predictor group after the others are controlled.
- **Signed hierarchical contributions:** complete contributions that retain negative components arising from suppression or collinearity.
- **Zero-truncated composition:** a display transformation that sets negative components to zero and recomposes the remaining contributions to sum to one.

This matters for how we describe the main figures. The zero-truncated bars divide the variation attributed to predictors within the fitted models; they are not percentages of all ecological change or estimates of pure unique variation. I also compared these summaries with a profile that excludes models containing negative individual components. The numerical allocations differ among profiles, but the aggregate ordering in the SPD-based analysis is retained.

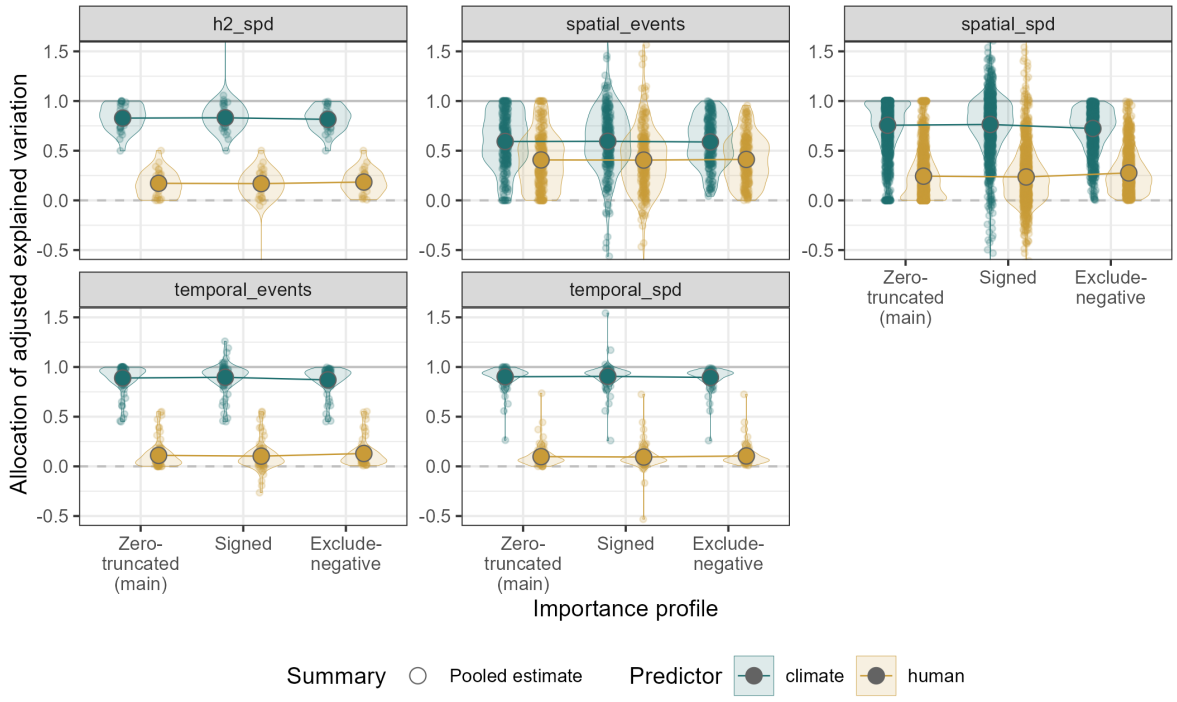


Figure 5: Comparison of signed, zero-truncated, and exclude-negative HVarPart profiles. These are alternative summaries of the same decompositions, not interchangeable effect estimates.

7 Showing model fit alongside predictor allocation

Once I had corrected the extraction, I also separated the question of model fit from the question of how the fitted variation is allocated. Of the 1,262 sequences considered for the initial spatial analysis, 1,156 have an estimable decomposition. Their mean total adjusted (R^2) is 0.352 [0.326, 0.374] (mean and region-block bootstrap 95% confidence interval).

The controlled spatial subset contains 1,098 sequences. Its mean adjusted (R^2) is 0.440 [0.426, 0.449], while its median and interquartile range are 0.450 [0.326, 0.567].

Fit varies substantially among sequences and is only weakly related to signed human allocation. A model can allocate most of its explained component to one predictor group while explaining little response variation overall; the reverse is also possible. I now report both quantities so that we do not read a large relative allocation as evidence of a strong overall fit.

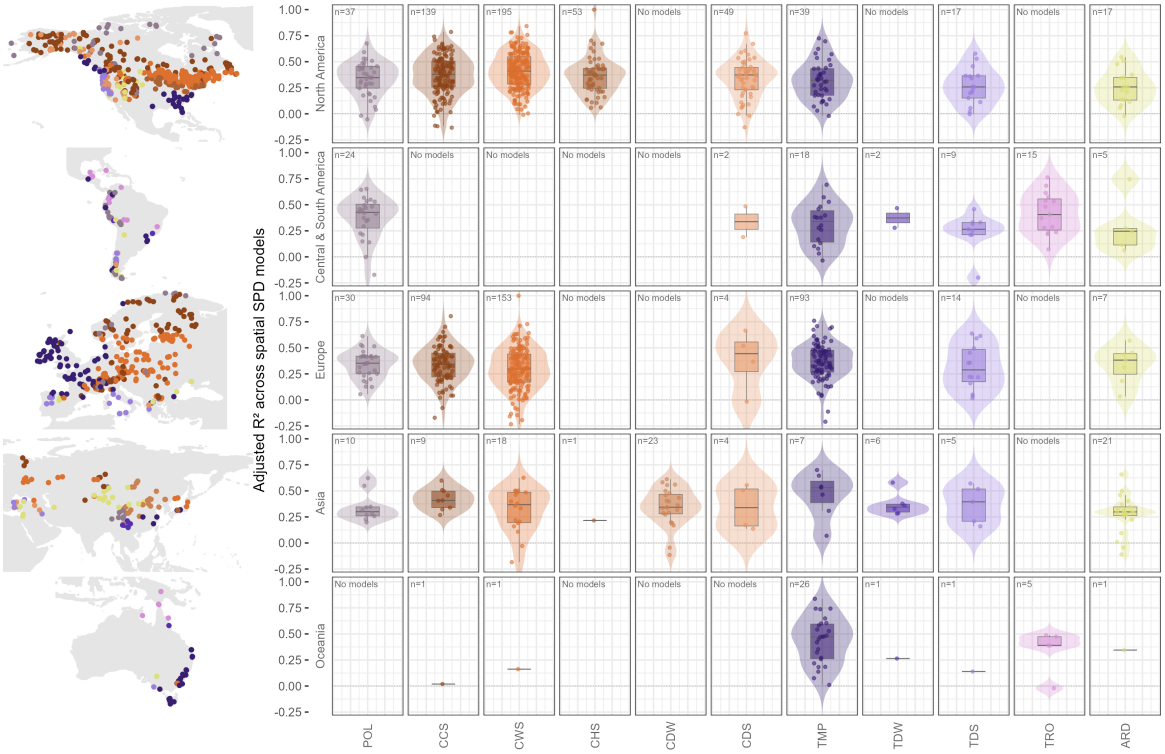


Figure 6: Distribution of total adjusted R-squared across spatial sequence models. Model fit is reported separately from the allocation of explained variation.

8 Adding time control to spatial H1

I next made the controls in the two H1 comparisons reciprocal: time is controlled within sequences for spatial H1, and spatial structure is controlled within time slices for temporal H1. For spatial H1, I fitted the model within each pollen sequence with time included as a protected control group. Human and climate contributions are then aggregated spatially using the existing distance-based Moran's eigenvector maps (dbMEM) procedure. This keeps temporal structure within a sequence from being silently attributed to either human activity or climate.

In this SPD-based analysis, climate has the larger zero-truncated contribution in 891 of 1,098 eligible sequences (81.1%). Some sequences and regions show stronger human allocations, but the overall distribution remains climate-dominant. This is the analysis behind Figure 3 in the revised manuscript.

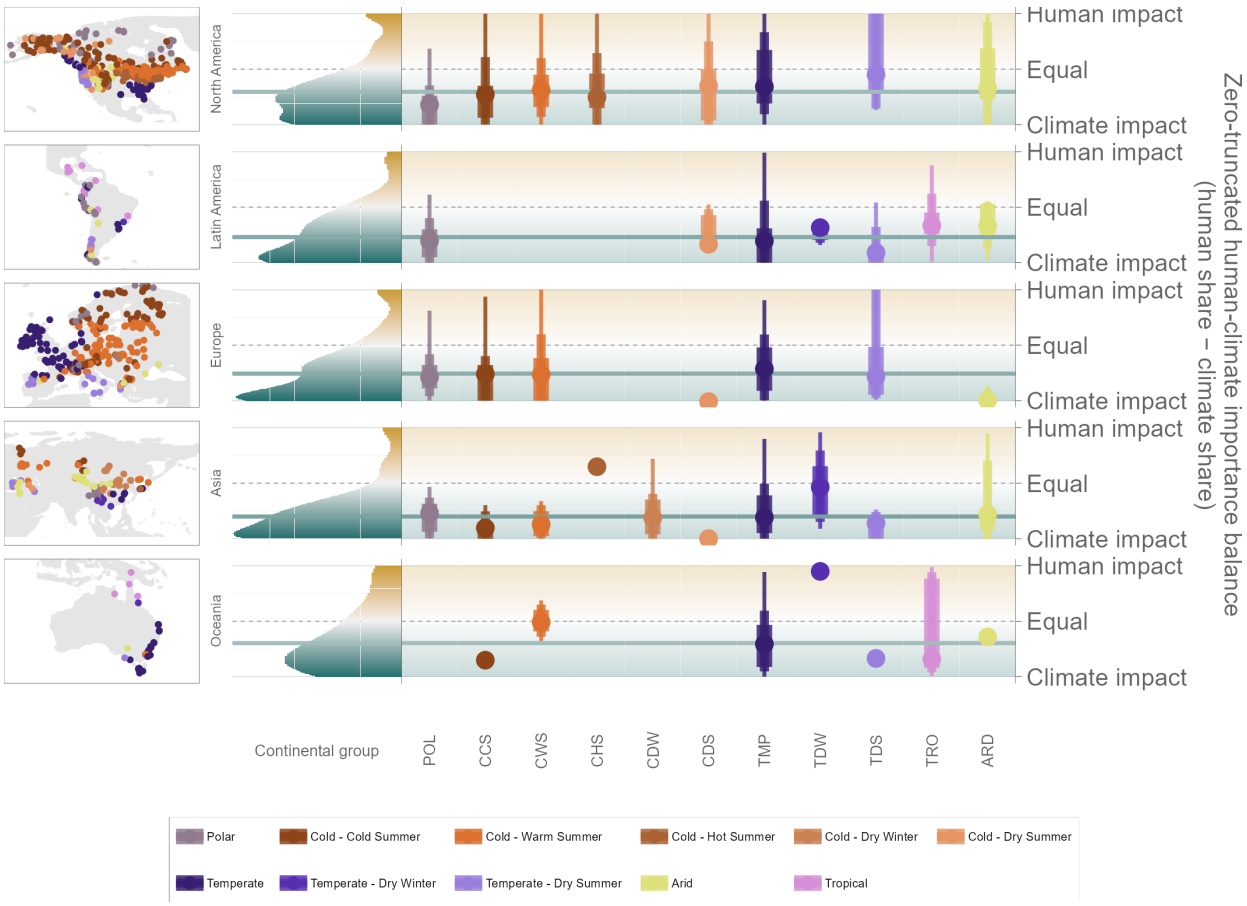


Figure 7: Revised manuscript Figure 3. Sequence-level human-minus-climate balance after controlling time within sequences and aggregating spatial pattern. Gold denotes greater human allocation and teal denotes greater climate allocation.

9 Adding spatial control to temporal H1

For temporal H1, I grouped sequences within 500-year continental-region time slices. Selected dbMEM variables represent spatial structure among the eligible sequences before I interpret the SPD and climate contributions. Where a combination cannot be estimated, I leave it missing rather than assign it a zero effect.

Climate has the larger zero-truncated contribution in 69 of 70 estimable continental-region-age comparisons. I also retained the signed and unique-fraction summaries, because negative components, shared variation, or weak unique signals can make the display composition incomplete. The figure below is the revised manuscript's Figure 4.

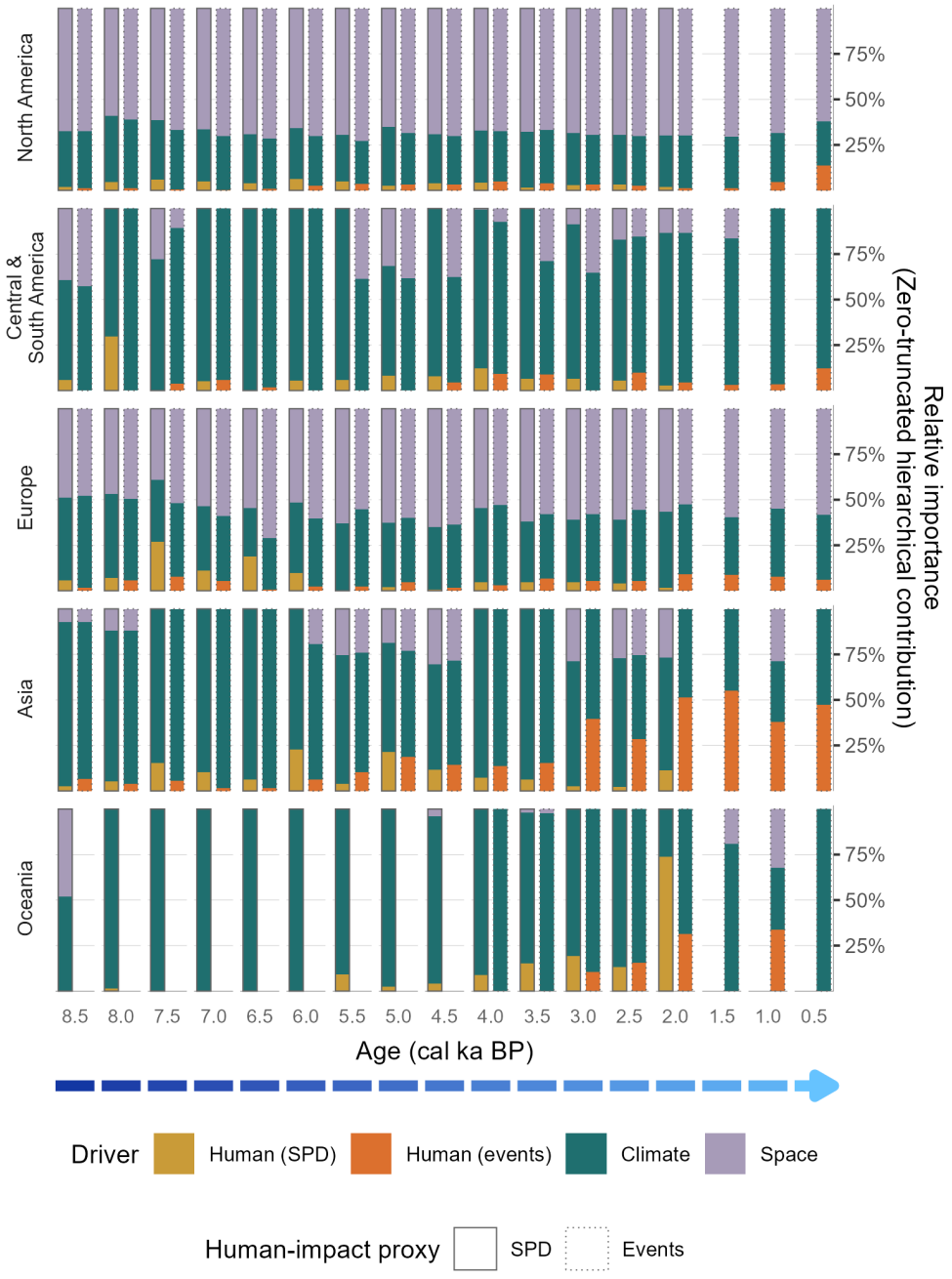


Figure 8: Revised manuscript Figure 4. Human, climate, and spatial allocations through time after spatial control within each continental-region time slice. Missing combinations are non-estimable.

10 Revisiting what H2 tells us

I also clarified the interpretation of H2, which asks whether human activity is the predominant predictor of changes in relationships among PAPs. Principal component analysis (PCA) and symmetric Procrustes distances summarise changes in collective PAP covariance between time slices within regions. Regional SPD and climate trajectories then enter a second HVarPart analysis.

The SPD-based H2 analysis attributes more of the broad change in PAP interrelationships to climate than to SPD in most estimable groups. The zero-truncated analysis-level climate allocation is 0.828.

I have been more careful about what we can infer from this ordination. It describes collective, multivariate change; it cannot identify which pollen taxon changed. Missing or short trajectories can also reflect incomplete estimability rather than ecological stability. H2 remains based on SPD alone, a point I will return to when we get to the final decision.

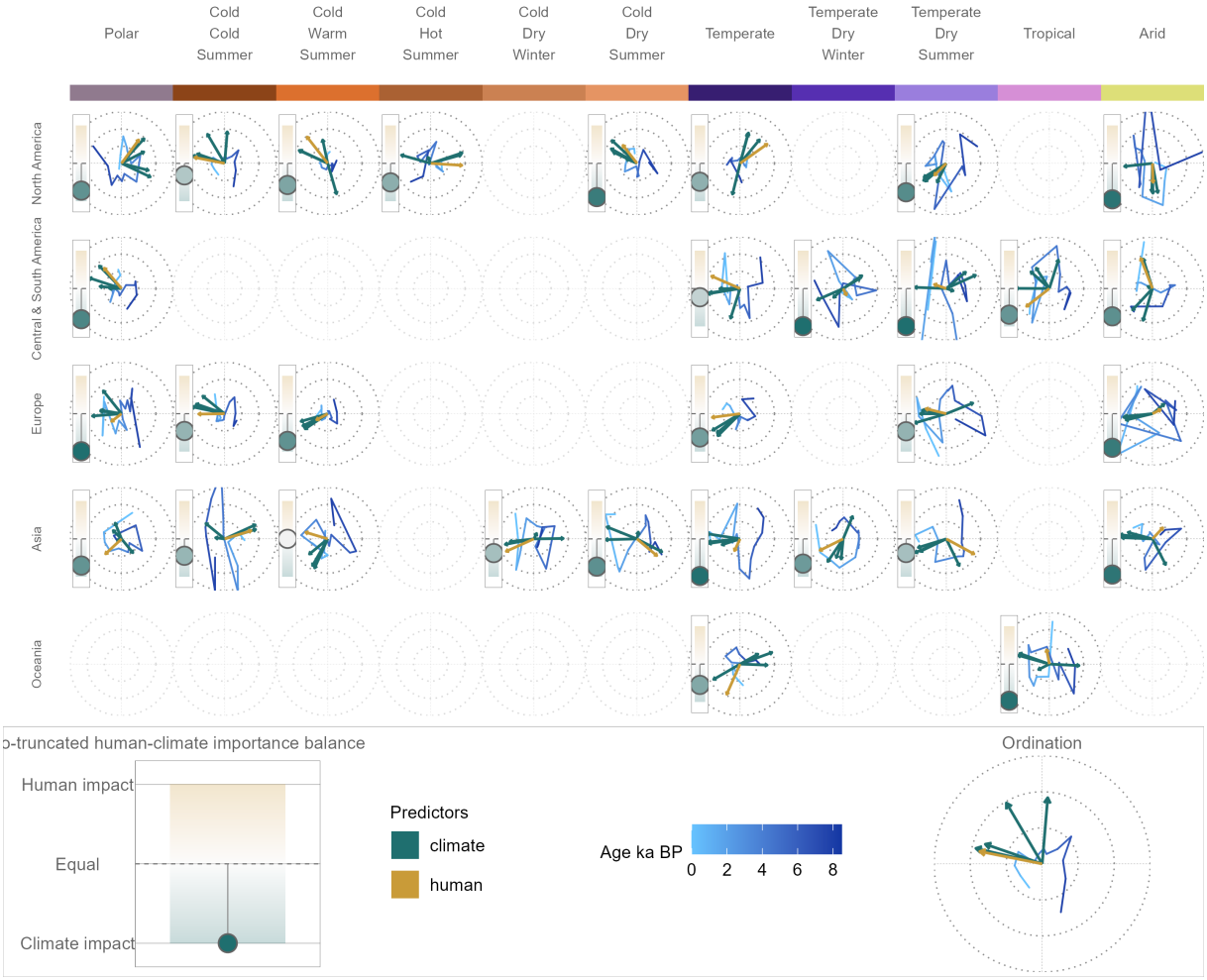


Figure 9: Revised manuscript H2 figure. Ordination trajectories describe changes in collective relationships among PAPs; predictor arrows are not taxon-level effects.

11 Checking dependence and uneven sampling

After adding the controls, I checked whether residual spatial structure, dense sampling, or particular geographical groups could still determine the result. The residual Moran diagnostics find positive spatial autocorrelation in 9 of 14 unfiltered comparisons and 4 of 14 residual comparisons after spatial control. Observed Moran's (I) ranges from -0.068 to 0.131 before control and from -0.061 to 0.054 afterwards.

I also ran spatial thinning and leave-one-group-out comparisons. Thinning retains a median of 361 sequence models at 250 km and 216 at 500 km across 100 replicates. Leaving out individual continental regions or regions generally retains a climate-dominant aggregate balance, while showing which omissions increase uncertainty. Together, these checks reduce the risk that dense sampling or one geographical group determines the result. They do not remove all dependence or establish causality, and I have kept that qualification in the manuscript.

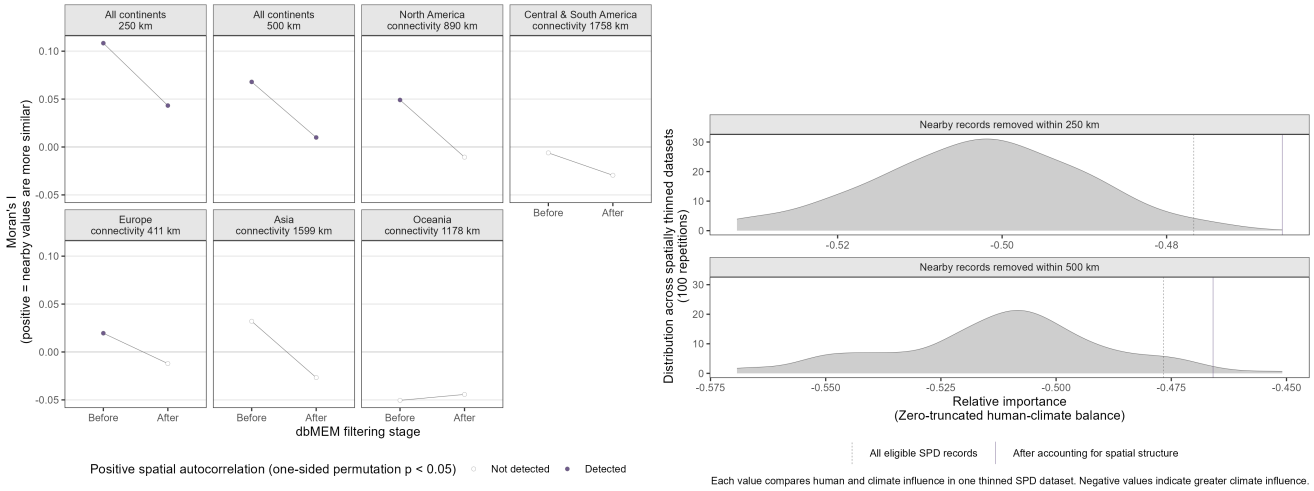


Figure 10: Residual Moran diagnostics before and after spatial control (left) and spatial-thinning sensitivity (right). Leave-one-group-out results are summarised in the text.

12 Testing the archaeological neighbourhood radius

Another choice I checked was the radius used to construct SPD around each pollen sequence. The main workflow uses archaeological dates within 250 km, with a 500 km fallback where the strict neighbourhood has insufficient dates. To separate the radius effect from differences in coverage, I compared strict 250 km and 500 km predictors on identical analytical units.

Among 999 matched spatial sequences, 963 have defined balances under both radii. The median 500-km-minus-250-km balance change is 0.025, and 94 sequences reverse their human-versus-climate ranking. At the temporal scale, 3 of 70 matched comparisons reverse ranking.

The fallback improves coverage without overturning the aggregate SPD-based conclusion, but it does change some individual estimates. I therefore describe this as a sensitivity to our archaeological neighbourhood choice. Neither radius is equivalent to the pollen-source area, so we cannot present it as a scale-matched archaeological comparison.

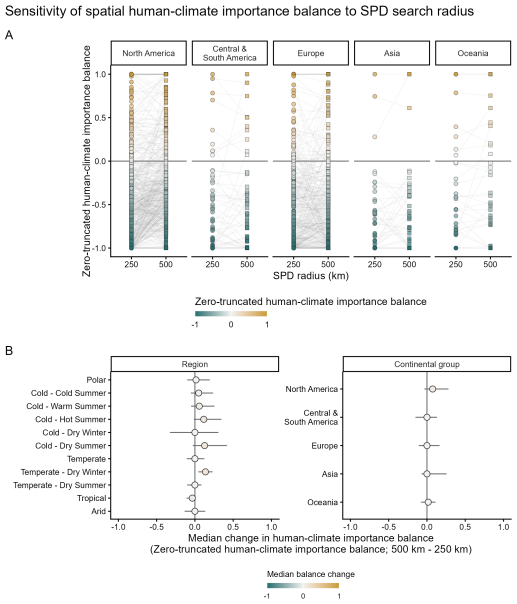


Figure 11: Matched SPD-radius sensitivity across spatial sequence models.

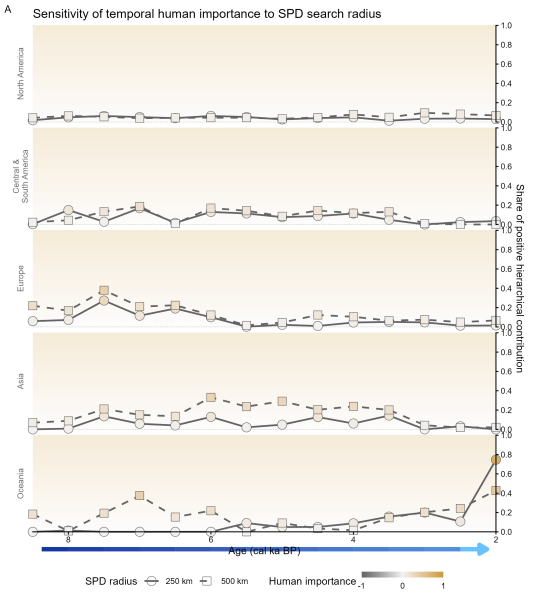


Figure 12: Matched SPD-radius sensitivity across continental-region time slices.

13 Comparing SPD with the expert-event histories

I then looked at the expert-screened pollen events as another source of evidence about human activity. These histories contain local information that SPD may miss, but they are coarser, geographically uneven, and incomplete. They are also partly circular: the event definitions and PAP responses can come from the same pollen sequence. For that reason, I used matched subsets with observed events rather than histories filled with default values.

The matched analysis contains 151 spatial sequences and 61 continental-region-age comparisons. Median spatial human-minus-climate balance is -0.018 for the combined SPD-plus-event model and -0.254 for the event-only model. Temporally, adding events changes median human allocation by 0.034, while substituting events changes it by -0.012.

The changes go in both directions, which tells us that human attribution depends partly on how we represent human activity. I have therefore kept the expert chronologies as complementary sensitivity evidence, rather than treating them as independent validation of SPD. SPD remains the primary human predictor in the revised manuscript.

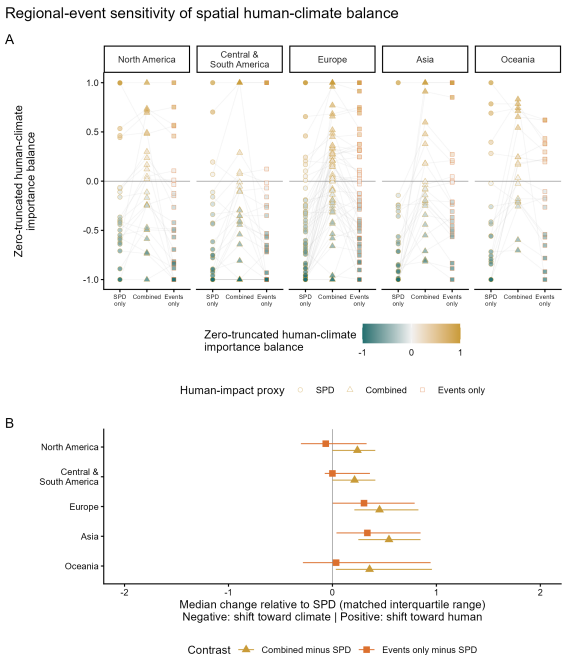


Figure 13: Matched expert-event sensitivity across spatial sequence models.

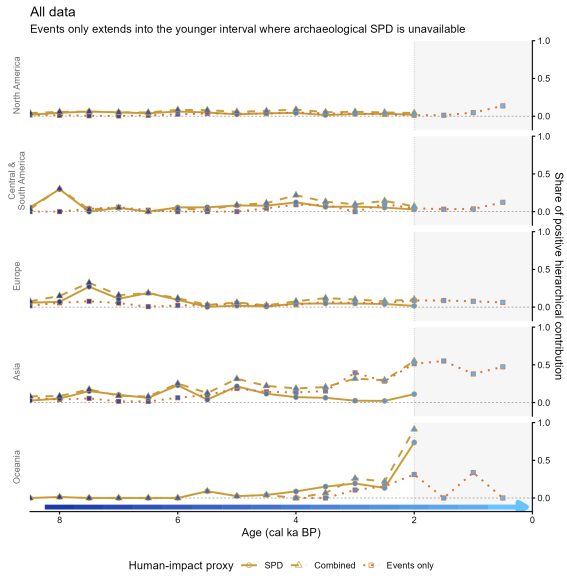


Figure 14: Matched expert-event sensitivity across continental-region time slices.

14 Bringing these changes back into the manuscript

By this point, I had a clearer basis for revising the manuscript and reviewer response. The SPD-based results support a more precise paper, but they do not support the broadest wording in the submitted version. I narrowed the scope and interpretation while keeping room for local human histories. The table below summarises how the framing changed.

Table 3: How I revised the framing of the submitted manuscript.

Element	Submitted framing	Current R1 framing
Scope	Holocene vegetation dynamics	Pollen assemblage dynamics from 8.5 to 2.0 ka BP.
H1	Human versus climate importance without reciprocal controls	Time-controlled spatial and spatially controlled temporal comparisons.
Importance	Relative importance presented as a simple ratio	Fit, unique fractions, signed commonality, and display composition kept distinct.
H2	Broad ecological interpretation of changing relationships	Collective PAP covariance with missingness and taxon-level limits explicit.
Human evidence	SPD and event evidence treated more comparably	SPD is primary; expert events are a complementary, partly circular sensitivity.
Conclusion	Broad claim about pre-industrial human importance	Scale- and interval-bounded statement that retains local human histories.

Table 4: What the supporting checks tell us about the SPD-based results and their limits.

Check	SPD-based finding	Qualification
Pollen-count coverage	Baseline retains the broadest usable sequence set.	Higher count targets sharply and unevenly reduce coverage; H1 was not refitted.
Reduced PAP response	Aggregate climate-over-human ordering persists.	Individual allocations change; PAPs are not independent replicates.
Extraction profiles	Ordering persists across all three summaries.	Signed, zero-truncated, and exclude-negative profiles answer different questions.
Spatial dependence	Controls reduce residual spatial structure.	Some residual dependence remains and limits causal interpretation.
Thinning and leave-out	Aggregate balance is generally retained.	Fewer sequences and particular omissions increase uncertainty.
SPD radius	Aggregate result is less sensitive than individual estimates.	Some sequences and time slices reverse ranking.
Expert events	SPD remains the primary analysis in R1.	Attribution varies with proxy definition; events are incomplete and partly circular.

I changed the title to name pollen assemblage dynamics and the 8.5–2.0 ka BP interval. In the abstract and Results, I now use the denominator appropriate to each analysis rather than treating the whole collection as one model population. In the Discussion, I distinguish broad-scale predominance from local human histories, remove unsupported tipping-point and uniformly destructive-impact claims, and spell out the limits arising from pollen counts, spatial mismatch, climate simulations, and proxy construction. The results do not establish that people were absent, locally unimportant, or ecologically benign.

I also rebuilt the reviewer response so that each concern can be followed through to the analysis and supporting evidence. That is where the SPD-based revision currently stands. There is still a central question, though: how well does SPD alone represent the human impact we want to discuss? The later work below is why I would like us to resolve that together before I take the manuscript further.

15 Final decision: should human impact include SPD, KK10, and HYDE?

15.1 Why I looked beyond SPD

After completing the SPD-based revision, I explored what happens if we represent human activity more broadly. I added KK10 land use and HYDE population as candidate predictors alongside SPD. This is exploratory work: I have not incorporated it in R1 or the reviewer response.

SPD primarily represents archaeological activity and demography inferred from radiocarbon-date densities. KK10 represents land-use fraction, while HYDE represents modelled population. Together they cover a broader human-impact construct than SPD alone, but they still have different assumptions, resolutions, and uncertainties.

I first checked how closely these proxies agree on the matched 2–8 ka comparison. The convergence dataset contains 15,353 observations from 1,181 pollen sequences and 13 ages. Rank agreement is positive overall: Kendall's tau is 0.87 for SPD versus KK10 and 0.69 for SPD versus HYDE. Agreement is less uniform among regions, which is one reason I used local predictor selection in the model comparison.

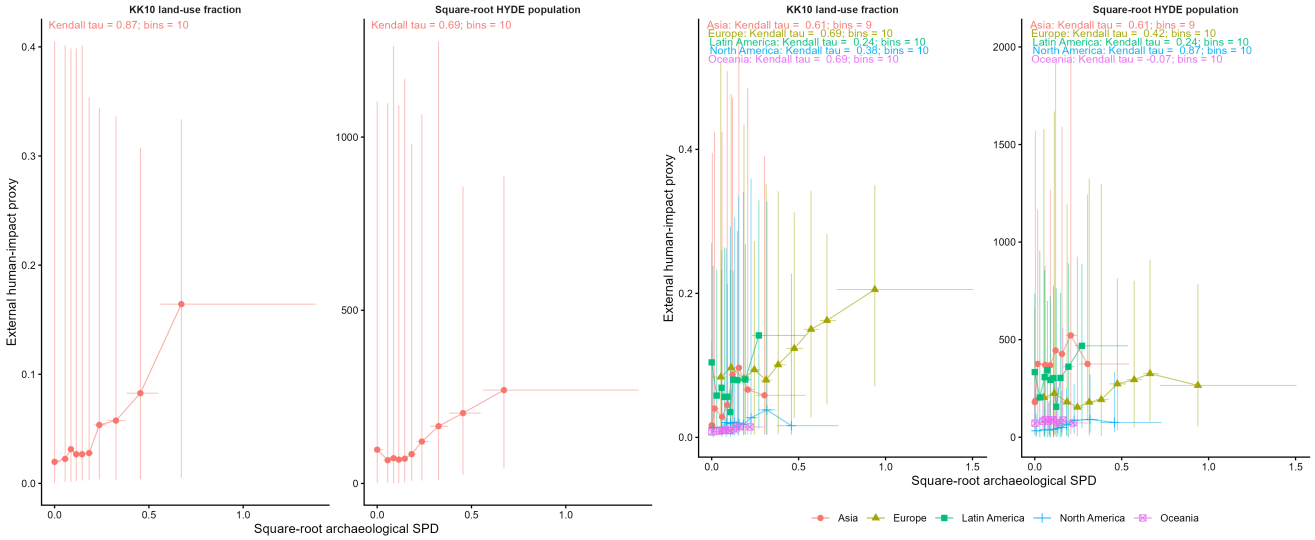


Figure 15: Agreement between SPD and the additional human proxies overall (left) and by continental region (right).

I also exported a separate spatial atlas showing SPD, KK10, and HYDE for the world and Europe at all thirteen matched time slices. I have not reproduced it here, but we can use it alongside these summaries when discussing differences in spatial support.

15.2 Making the model comparison fair

Simply adding predictors would mix several changes together, so I set up a matched comparison. The exploratory model uses square-root SPD, KK10 fraction, and square-root HYDE as candidate human predictors. The square-root transformations reduce extreme right-skew while retaining ordering; KK10 remains on its natural fractional scale.

Within each sequence or continental-region-age unit, I removed redundant predictors separately within the human and climate blocks using ,($|r| = 0.8$) and a variance inflation factor (VIF) threshold of 5. The human preference order is square-root SPD, KK10, then square-root HYDE; the climate order remains annual temperature, coldest-month temperature, warm-season precipitation, then cold-season precipitation. I protected the time and dbMEM controls and added them after selecting the human and climate predictors. I diagnosed cross-block correlation without using it for automatic removal, because shared human-climate variation is part of the partitioning question.

The retained predictors can therefore differ among analytical units. This reduces redundancy within each block, but does not establish that a retained variable is causal or that this is the uniquely correct way to represent human impact.

I compared three models:

1. **The current SPD model:** square-root SPD, all four climate predictors used in R1, and the 8.5–2.0 ka BP interval.
2. **The matched SPD bridge:** square-root SPD, the joint model's locally selected climate set, and the same matched 8–2 ka observations as the joint model.
3. **The filtered joint human block:** square-root SPD, KK10, and square-root HYDE after local human filtering, with the same climate selection and cohort as the bridge.

Comparing the current SPD model directly with the joint model combines changes in interval, observations, climate selection, and human representation. The bridge-to-joint comparison is cleaner: the observations and climate predictors stay the same, so it isolates the change in the human block. The figure below shows the spatial result for that matched SPD bridge.

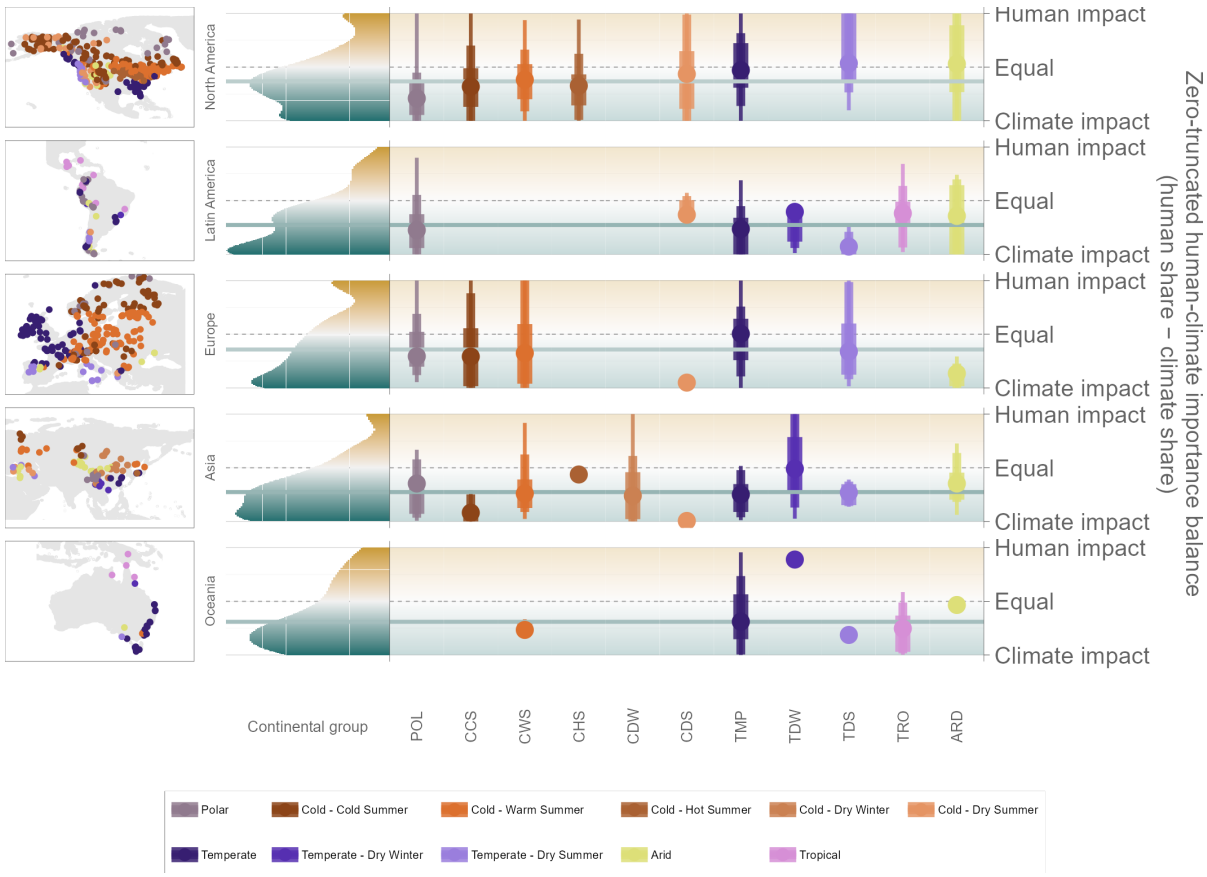


Figure 16: Figure 3 analogue for the matched SPD bridge. Climate remains dominant on the harmonised observations and locally selected climate sets.

15.3 What changed in spatial H1

This is where the additional proxies make a substantial difference. The exact three-model common spatial subset contains 1,069 sequences. On that subset, the current SPD model is climate-dominant in 869 sequences, and the matched SPD bridge remains climate-dominant in 764.

The filtered joint model is instead human-dominant in 692 sequences and climate-dominant in 376, with 1 tie. The median zero-truncated human-minus-climate balance moves from -0.35 in the bridge to 0.18 in the joint model. Moving from bridge to joint changes 387 sequences from climate- to human-dominant, with no corresponding human-to-climate reversal.

So the spatial ordering reverses even when I hold the matched observations and climate selection fixed. This is not just a consequence of comparing different subsets. The figure below is the joint-model counterpart to Figure 3, but it is not yet the manuscript figure.

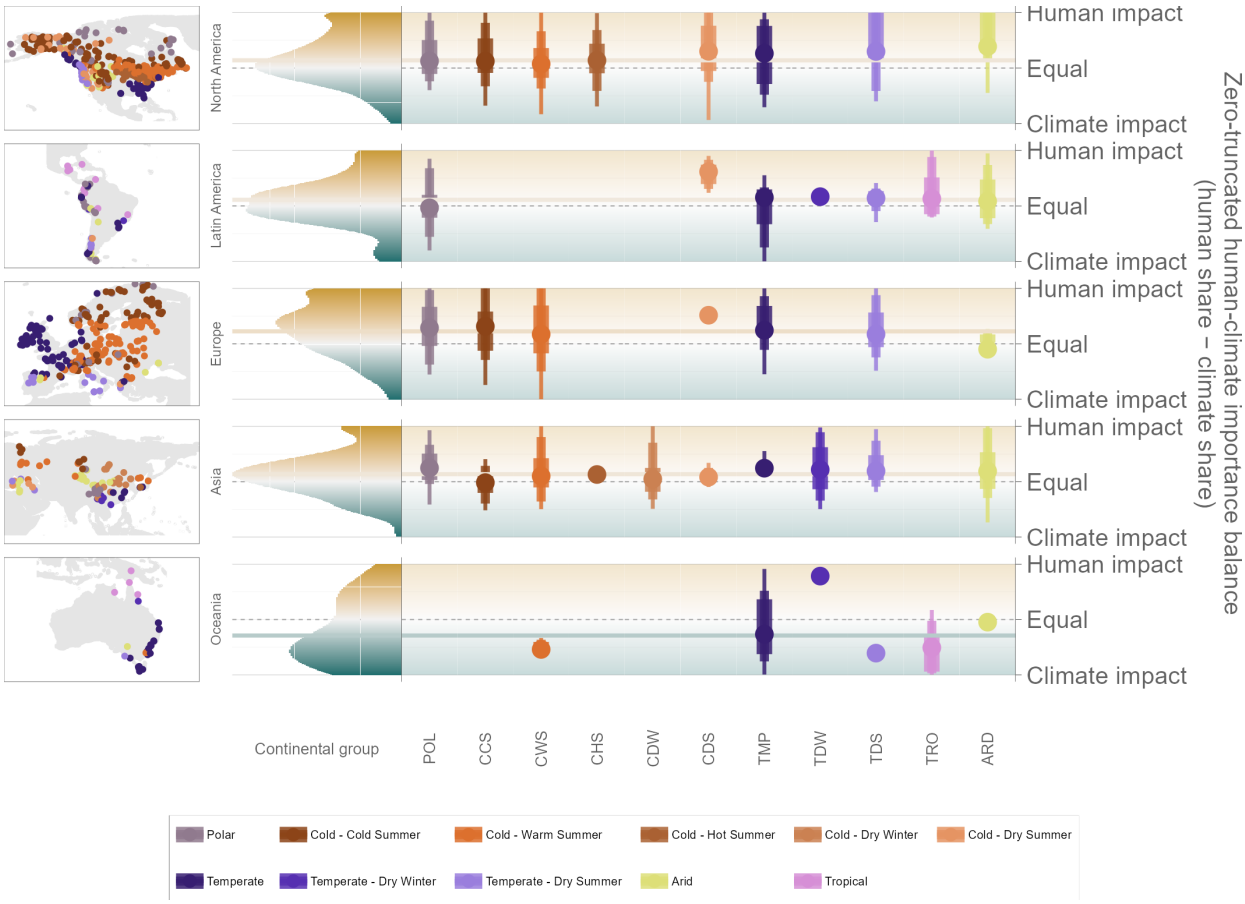


Figure 17: Figure 3 analogue for the filtered joint human block on the exact common subset. This is exploratory evidence and is not the current manuscript Figure 3.

15.4 What changed in temporal H1

I then made the same three-model comparison for temporal H1. All three models are compared across the same 65 continental-region-age units. The current SPD model is climate-dominant in 64 units and the matched SPD bridge in 65. Under the filtered joint model, climate remains larger in 47 units and humans are larger in 18.

Median human allocation increases from 6% in the bridge to 26% in the joint model. The bridge-to-joint contrast produces 18 climate-to-human reversals. Here the conclusion weakens substantially, but climate remains dominant overall. The spatial and temporal results therefore do not support one simple replacement headline.

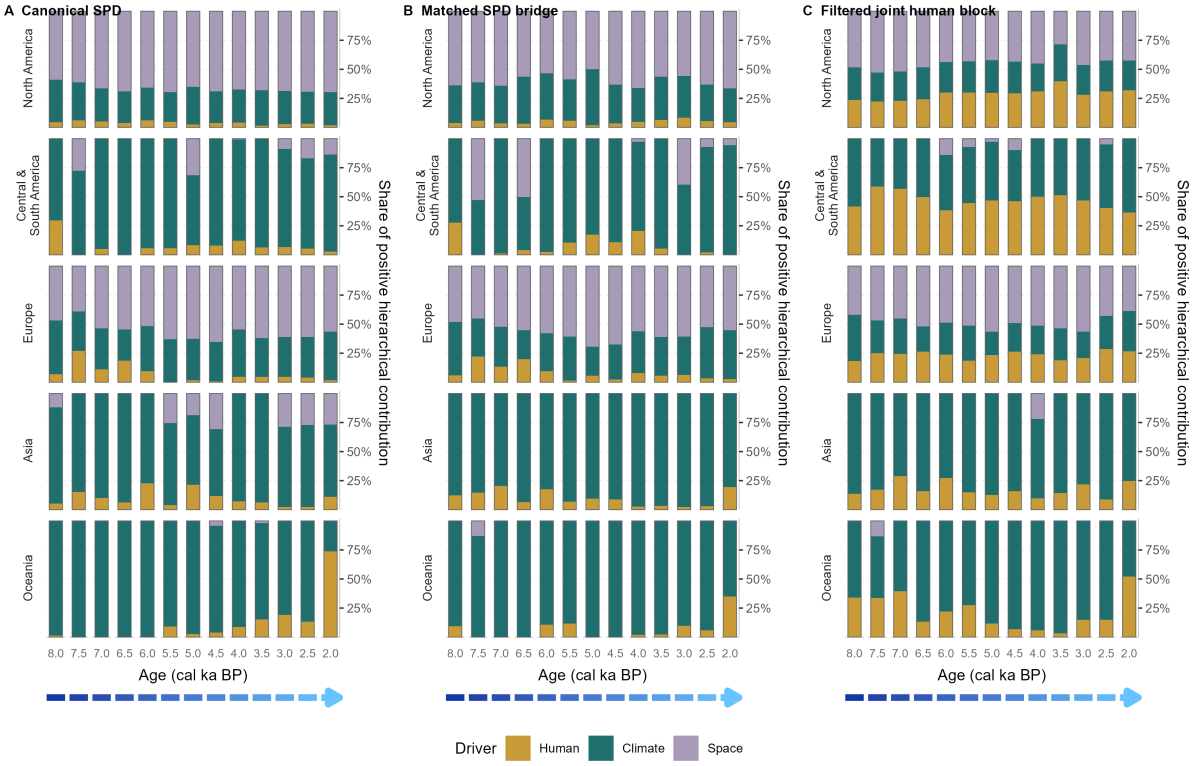


Figure 18: Figure 4 comparison on the exact common subset: current SPD model, matched SPD bridge, and filtered joint human block. Every displayed stack sums to one after zero truncation.

15.5 Checking fit and what each local model retains

Before drawing a manuscript conclusion from those changes, I also checked model fit and which predictors survive the filtering. Adding the filtered joint human block changes both allocation and fit. Across the matched sequence comparison, median total adjusted (R^2) changes from 0.452 to 0.480. Pure human adjusted (R^2) generally increases, while the pure climate and time fractions often decrease, reflecting both the broader human block and changes in shared variation.

Predictor-retention frequencies vary among continental regions and regions. I think that variation is useful to show, but it also means we need to be careful about saying that every local model represents exactly the same human construct.

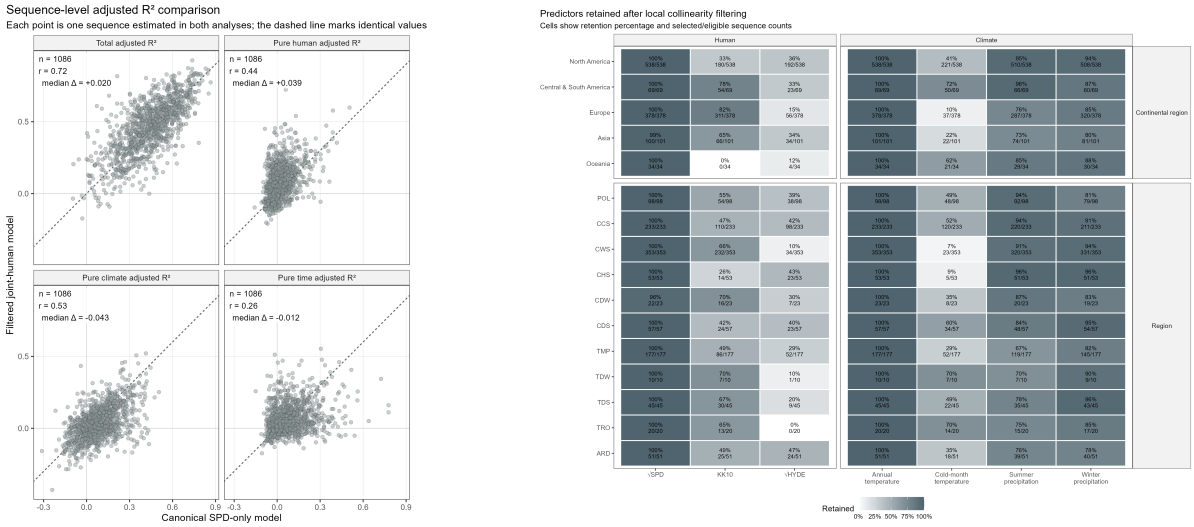


Figure 19: Sequence-level adjusted R-squared comparison (left) and local predictor retention by continental region and region (right).

15.6 Where I need your input

Taken together, these results mean that our claim that climate generally outweighs human impact is not robust to this defensible expansion of the human predictor block. That does not establish that the joint model is the only correct specification. It does mean that we need to agree what we want to represent by human impact and how openly we show the dependence of the result on that choice.

Including KK10 and HYDE would require a substantial rewrite of the manuscript, rather than simply adding another sensitivity figure. I would need to revisit the framing, Results, title, abstract, Discussion, conclusions, and reviewer response. There is also unfinished analytical work: H2 still uses SPD alone, and I have not refitted it with KK10 and HYDE. If we make the joint model primary, or treat both specifications as co-primary, we would need a harmonised H2 comparison before revising the paper-wide conclusions.

15.6.1 The three options I see

1. **Keep SPD as the primary analysis.** Retain the current Figures 3–4, show the joint model as a sensitivity, and substantially soften the climate-dominance headline. This keeps the primary H1 analysis aligned with the current SPD-only H2, but leaves us with a narrower representation of human activity.
2. **Make the joint model the primary H1 analysis.** The spatial result becomes human-dominant, while the temporal result remains mixed and climate-dominant overall. I would need to refit H2 with the harmonised human proxies before rewriting the overall interpretation.
3. **Present both specifications as co-primary.** We would make the dependence on our representation of human impact central to the paper, and harmonise H2 across the two specifications.

15.6.2 What I would like us to agree on

The first question is whether SPD alone is the human-impact construct we want to defend, or whether archaeological activity, land use, and modelled population together are closer to what we mean. From there, we can decide how prominently to present both specifications and whether the resulting scale- and specification-dependent conclusion should become the main message. We also need to settle the extent of the H2 work before I restructure the manuscript and reviewer response.

I have not chosen between these options. Vivian and Suzette, I would really value your thoughts on which approach best reflects the evidence and gives us the clearest, most transparent response to the reviewers. What I would like us to come away with is an agreed representation of human impact, a provisional direction for the manuscript, and a decision on the H2 refit before I take on the larger rewrite.